In-Country Working Session: FASTR Implementation & RMNCAH-N Service Monitoring Analysis

January 27-30, 2026 | Lusaka

GFF FASTR Team

FASTR
FREQUENT ASSESSMENTS & SYSTEMS TOOLS FOR RESILIENCE

Agenda

Day 1 -- Laying the Foundation: Introducing FASTR and Configuring the Analytics Platform

Time	Agenda	Facilitator/Presenter
Opening Session		
08:30-09:00	Participant registration	MoH team
09:00-09:10	Welcome and opening remarks	MoH team
09:10-09:20	Icebreakers/Introductions	MoH team
09:20-09:35	Overview of agenda, workshop objectives	GFF FASTR team
Session 1: Overview of the FASTR approach		
09:35-10:30	Overview: FASTR Approaches	GFF FASTR team
Session 2: HMIS data extraction		
10:30-11:30	Data extraction: Rationale and methods	GFF FASTR team
Session 3: Introduction to the FASTR analytics platform		
11:30-12:30	Introduction to the FASTR analytics platform	GFF FASTR team
12:30-14:00	*Lunch Break*	
14:00-14:30	Getting participants into the platform	GFF FASTR team
Session 4: Configuring the FASTR analytics platform		
14:30-16:30	Configuring the analysis platform	GFF FASTR team

Agenda

Day 2 -- Building the Analysis: Applying FASTR Methods and Generating Outputs

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 2 agenda	GFF FASTR team
Session 5: Overview of FASTR methods and analytical outputs		
09:15-10:15	Data quality, service utilization, coverage	GFF FASTR team
Session 6: Creating a project		
10:15-11:15	Project creation and settings	GFF FASTR team
Session 7: Creating visualizations		
11:15-12:30	Creating and editing visualizations	GFF FASTR team
12:30-14:00	*Lunch Break*	
Session 8: Creating reports		
14:30-16:30	Practice creating and editing reports	GFF FASTR team

Agenda

Day 3 -- From Analysis to Action: Interpreting Results and Using FASTR for Decision-Making

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 3 agenda	GFF FASTR team
Session 8: Interpretation of visualizations		
09:15-10:15	Approaches to support interpretation	GFF FASTR team
Session 9: Creating a Q4 2025 report		
10:15-12:30	Creating short and long reports with country context	GFF FASTR team
12:30-14:00	*Lunch Break*	
Session 9 (cont'd): Creating a Q4 2025 report		
14:00-15:00	Continue report creation with country context	GFF FASTR team
Session 10: Presenting reports		
14:30-15:30	Present reports, group feedback	GFF FASTR team

Agenda

Day 4 -- Designing the Health Facility Assessment

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 4 agenda	GFF FASTR team
Session 12: Overview of FASTR HFA phone survey		
09:15-10:15	HFA overview and questionnaire adaptation guidelines	GFF FASTR team
Session 13: Questionnaire adaptation to the Zambian context		
10:15-12:30	Review questionnaire + hands-on adaptation	GFF FASTR team
12:30-14:00	*Lunch Break*	
Session 14: Questionnaire adaptation (cont'd)		
14:00-15:00	Continue questionnaire adaptation (in groups)	GFF FASTR team
Session 15: Discussion: HFA adapted questionnaire		
14:30-15:30	Discuss adapted questionnaire in plenary	GFF FASTR team
Session 16: Discussion: HFA priorities and data use		
15:30-16:30	HFA priorities and data use case in Zambia	GFF FASTR team
Session 17: Action planning and wrap-up		
16:30-17:00	Key messages and wrap-up	GFF FASTR team

Workshop Objectives

- Prepare MoH for participation in the multi-country workshop
- Strengthen capacity for disruption analysis and data extraction
- Configure the FASTR analytics platform for Zambia
- Produce the first quarterly report
- Adapt the phone survey questionnaire to the Zambian context
- Use data downloader tools for DHIS2

Scope of Work

Disruption Analysis Activities

- Data extraction and configuration of analytics platform
- Produce first quarterly report
- Review results and contextualize findings (with relevant program teams)

Phone Survey Activities

- Adaptation of questionnaire (with program input)

Capacity Building

- Training on data downloader for all with DHIS2 access
- Hybrid format: face-to-face and online

Expected Outputs

- Adapted phone survey questionnaire
- First quarterly report draft
- Trained MoH team on data downloader tools
- Roadmap for participation in Abuja workshop

Session 1: Overview of the FASTR approach

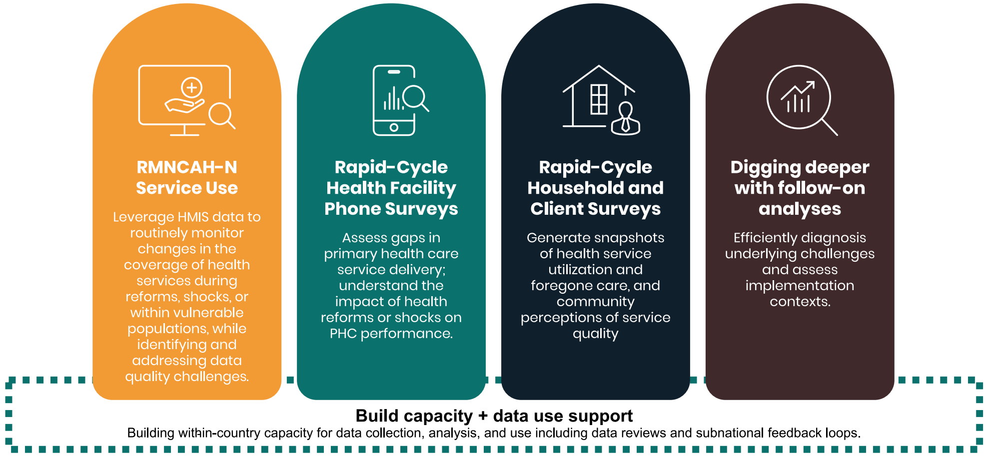
Introduction to FASTR

The Global Financing Facility (GFF) supports country-led efforts to strengthen the use of timely data for decision-making, with the goal of improving primary healthcare (PHC) performance and RMNCAH-N outcomes.

Frequent Assessments and Health System Tools for Resilience (FASTR) is the GFF's rapid-cycle analytics framework for monitoring health system performance using high-frequency data.

FASTR brings together four complementary technical approaches:

1. Routine HMIS data analysis
2. Health facility phone surveys
3. High-frequency household phone surveys
4. Follow-on, problem-driven analyses



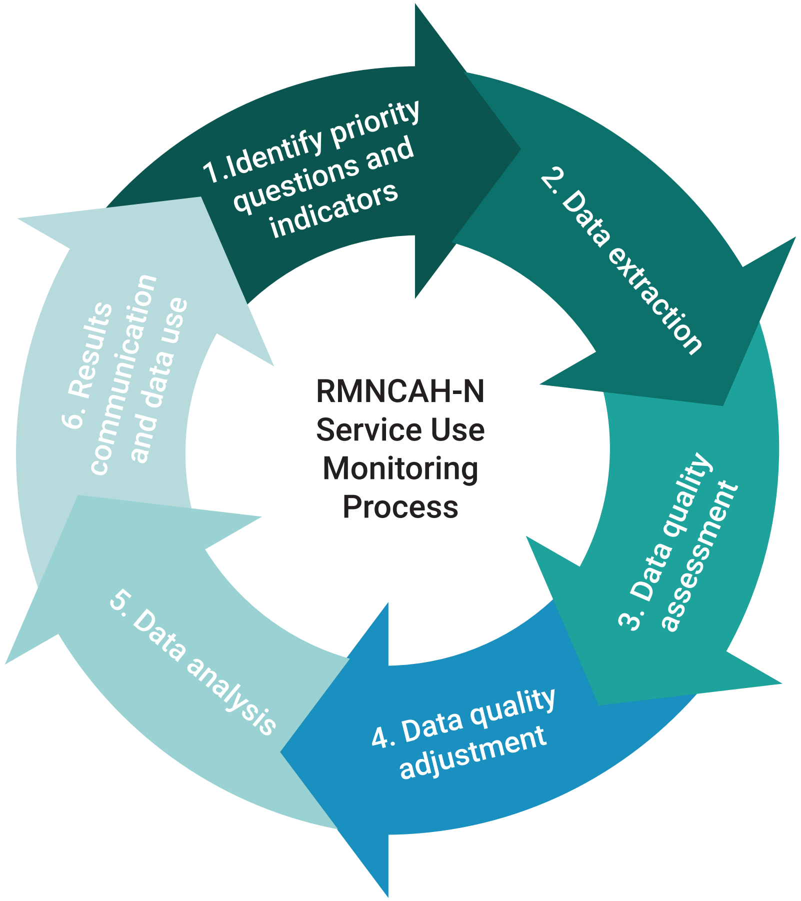
RMNCAH-N service use monitoring

Rapid-cycle approaches using routine health management information system (HMIS) data can be used to:

- **Evaluate the quality of HMIS data** both at the national and sub-national levels to help address gaps in quality and completeness
- **Measure monthly changes** in the utilization of critical health services, enabling a rapid response to challenges and the opportunity to gain insights on the progress of reforms
- **Compare service coverage trends** with country targets, facilitating the continuous monitoring of RMNCAH-N progress

How does it work?

Through collaboration with Ministries of Health, the GFF assists countries in developing and reviewing regular analyses of HMIS data, focusing on specific priority health service indicators linked to national health care reforms, as well as GFF and World Bank investments.



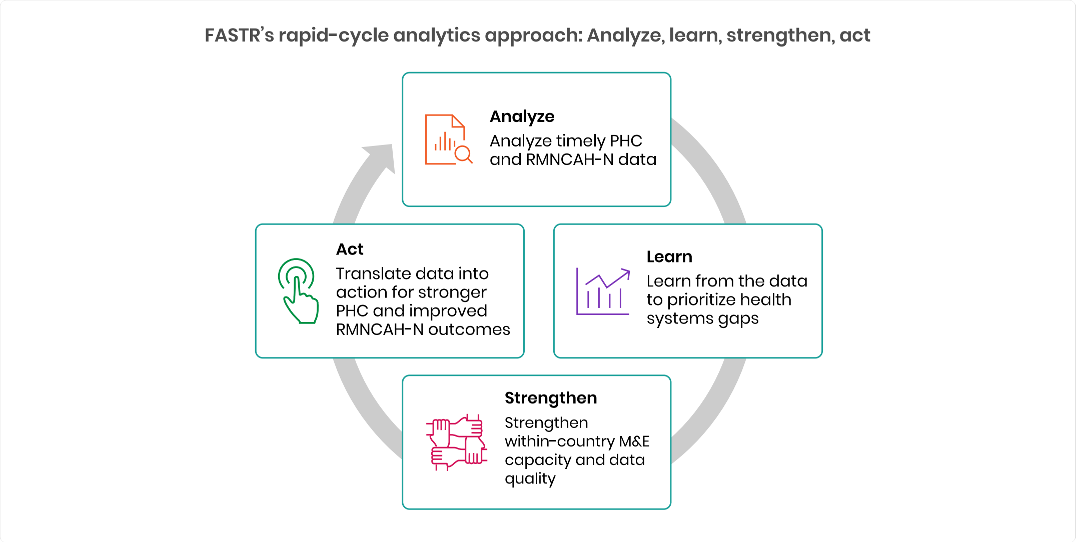
Why rapid-cycle analytics?

Routine health information systems are a critical source of data, but they are often underused due to concerns about data quality and long delays between data collection and analysis. Traditional household and facility surveys, while essential, are resource-intensive and infrequent.

FASTR's rapid-cycle analytics address this gap by providing:

- Timely insights aligned with country decision cycles
- Continuous learning rather than one-off assessments
- Direct feedback loops between data, analysis, and action

During the COVID-19 pandemic, this approach was applied in over 20 countries to monitor disruptions to essential RMNCAH-N services and inform response and recovery planning.



Focus of the analysis

Core indicators

FASTR prioritizes a core set of RMNCAH-N indicators that:

- Represent key service delivery contacts across the continuum of care
- Have relatively high reporting completeness and volumes
- Serve as proxies for broader service delivery performance

Outpatient consultations are included as a proxy for overall health service use. The indicator set can be expanded to reflect country-specific priorities.

Core data quality metrics

Analysis is anchored in a standardized set of data quality metrics, including:

- Reporting completeness
- Extreme value (outlier) detection
- Consistency across related indicators

These metrics are summarized into an overall data quality score to support interpretation and comparison across areas.

Session 2: HMIS data extraction

Why extract data from DHIS2?

Data quality adjustment

The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

The FASTR methodology includes specific approaches to:

- Identify and adjust for outliers
- Adjust for incomplete reporting
- Apply consistent data quality metrics

These adjustments require processing that cannot be done within DHIS2's native analytics.

Why extract data from DHIS2?

Analysis complexity

The FASTR approach uses more advanced statistical methods, such as regression analysis, which are not available in DHIS2. While DHIS2 can plot trends over time using raw data, FASTR can go further by:

- Identifying significant increases or decreases in service volume
- Adjusting for data quality issues
- Accounting for expected seasonal variations
- Comparing key periods, such as before and after a reform

The choice between DHIS2 and the FASTR approach should be guided by the specific purpose of your analysis.

Data format and granularity

Data should be downloaded for each **indicator of interest**, at **facility level**, and **monthly** for the **period of interest**.

- Data should be saved in **long format** meaning each row represents a single observation or measurement
- Data should be saved in **.csv format** and can be saved in either a single .csv file or multiple .csv files

Why monthly facility level data?

We want to use the most granular data we have access to in order to make more fine tuned assessments for data quality. Using monthly facility level data allows us to conduct the most robust analysis.

Key variables

The data extracted should include the following required elements:

Element	Description
Org units	Organizational unit identifier
Period	Time period of the data
Indicator name	Name of the indicator
Total/count	The aggregated value

How much data?

Initial FASTR analysis

- Download approximately **five years** of historical data
- Exact period depends on data availability and consistency in indicator definitions

Routine update to FASTR analysis

- Download new data covering the most recent months not previously included (usually **three months** for quarterly implementation)
- Include the **three preceding months** as recent data is often subject to changes due to late reporting or data quality adjustments

Data extraction tools

We offer two tools for bulk DHIS2 data extraction:

API Script (Google Colab)

- Input login credentials, specify timeframes, indicators, and administrative levels
- Download data as a .csv file

Data Downloader

- More intuitive, streamlined interface
- Recommended for most users

Both tools enable efficient data extraction, and we provide training resources to support their use.

DHIS2 Data Downloader

The Data Downloader is a desktop application for extracting data from DHIS2.

Key features:

- Connect to any DHIS2 instance
- Browse and select data elements and indicators
- Download facility-level data in CSV format
- Maintain download history

Download from GitHub:

https://github.com/worldbank/DHIS2-Downloader/releases/

Facilitator will demonstrate the Data Downloader

Session 3: Introduction to the FASTR analytics platform

Introduction to the FASTR Analytics Platform

The FASTR analytics platform is a web-based tool for data quality assessment, adjustment, and analysis of routine health data.

Key features:

- Upload and analyze data from DHIS2 and other sources
- Built-in statistical methods for data quality adjustment
- User-friendly interface for running analyses
- Flexible visualization and export options

In this session, we will provide a conceptual walkthrough of the platform and its capabilities.

Live Demo: Platform Access & Roles

In this demo, we will:

- Navigate to the FASTR platform
- Explore user roles: Administrator, Editor, Viewer
- Review user management and permissions
- Understand the workflow for uploading data and making analytical decisions

Facilitator will demonstrate in the live platform

Lunch Break

90 minutes

Back at 14:00

Session 4: Configuring the FASTR analytics platform

Activity: Setting Up Admin Areas

In this hands-on session, we will configure:

- Admin areas (regions, districts)
- Facility structure
- Indicator definitions

Participants will work directly in the platform

Activity: Importing Data

In this hands-on session, we will:

- Review data format requirements
- Walk through the import process
- Handle validation and error checking

Participants will import their country's data

Activity: Installing and Running Modules

In this hands-on session, we will:

- Review available analysis modules
- Install required modules
- Run initial analyses

Participants will configure and run modules on their data

Day 2

Recap

On Day 1, we covered:

- The FASTR approach and rapid-cycle analytics methodology
- Data extraction from DHIS2 using the Data Downloader
- Introduction to the FASTR Analytics Platform
- Platform configuration: admin areas, data import, modules

Day 2 Focus

Today we will:

- Explore FASTR methods for data quality and analysis
- Create and configure projects
- Build visualizations
- Generate reports

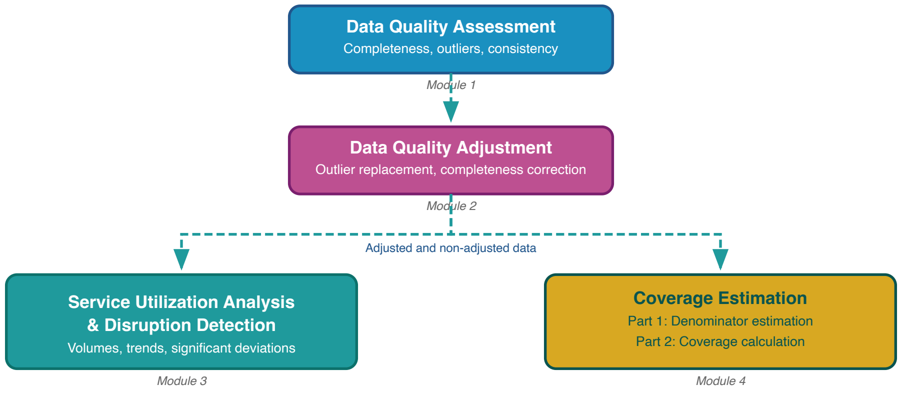
Session 5: Overview of FASTR methods and analytical outputs

FASTR analytical pipeline

The FASTR analysis follows a sequential workflow where each step builds on the previous:

1. **Assess data quality** - Identify issues with completeness, outliers, and consistency
2. **Adjust for quality issues** - Apply corrections to improve data reliability
3. **Analyze adjusted data** - Generate service utilization and coverage estimates

Each step must be completed before moving to the next.



Data quality assessment

Evaluating the reliability of routine health information system data

Rationale for data quality assessment

Challenge: Routine health facility data may contain quality limitations:

- Reported values may fall outside plausible ranges
- Reporting gaps affect data completeness
- Inconsistencies exist between related indicators

Implications: Data quality limitations affect decision-making

- Inaccurate assessments of service delivery trends
- Misidentification of areas requiring intervention
- Suboptimal resource allocation

Objectives of data quality assessment

Objective 1: Enable analytical adjustment

Systematic data quality assessment supports the application of targeted adjustments, enhancing the utility of HMIS data for evidence-based decision-making.

Objective 2: Monitor data quality trends

Data quality assessment enables ongoing monitoring to:

- Inform indicator selection based on quality profiles across the HMIS
- Guide targeted data quality interventions and supportive supervision in areas with weaker data quality
- Evaluate the effectiveness of data quality improvement initiatives over time

Core dimensions of data quality

1. Completeness

Are health facilities submitting reports consistently?

2. Outlier prevalence

Are reported values within plausible ranges?

3. Internal consistency

Do related indicators demonstrate expected relationships?

These three dimensions provide a comprehensive assessment of data reliability for analytical purposes.

Assessing reporting completeness

Definition:

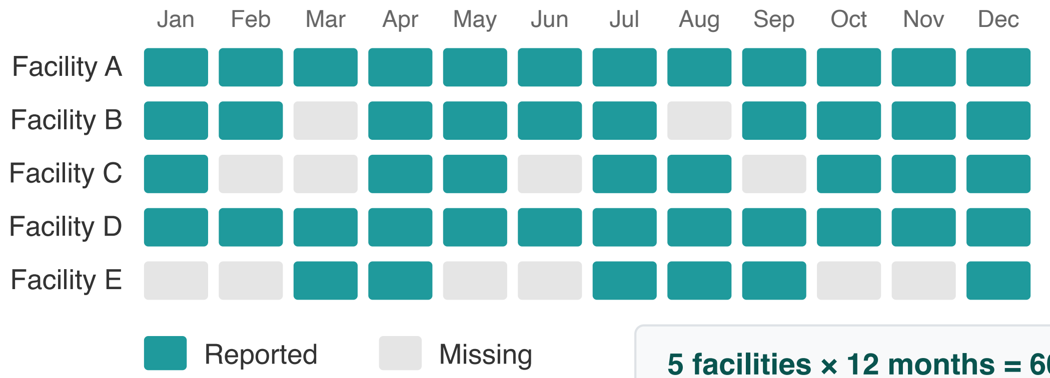
The proportion of expected facility reports that were submitted within a given period.

Significance:

- Incomplete reporting limits the representativeness of aggregate data
- Apparent declines in service volumes may reflect reporting gaps rather than actual reductions

Region A: Indicator Completeness

5 health facilities reporting monthly on ANC1



5 facilities × 12 months = 60 expected
48 reported → 80% completeness

Interpreting completeness rates

Reference benchmarks:

- 90% and above: High completeness
- 80-89%: Moderate completeness
- Below 80%: Substantial reporting gaps

Considerations: Complete reporting does not capture services delivered outside the formal health system or by facilities not included in the HMIS.

Key analytical questions: Are completeness rates improving over time? Which geographic areas or facility types demonstrate lower completeness?

Completeness: FASTR output

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

90% or above
80% to 89%
Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

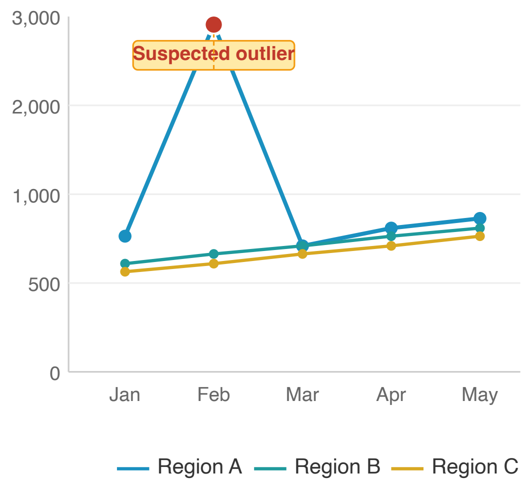
Identifying implausible values

Illustration:

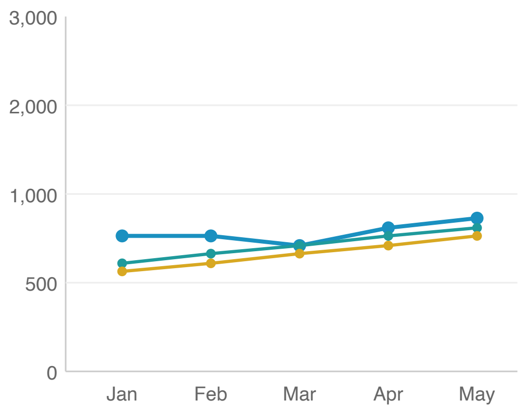
Region A displays an anomalous increase in February that substantially exceeds values reported by other regions.

This pattern is indicative of a data entry error. Following adjustment, all regions demonstrate consistent gradual trends.

With outliers included
(Region A shows spike in February)



Without outliers
(Regions show similar trends)



Outliers can distort the true picture of service trends

Outlier detection methodology

Outliers are identified through analysis of within-facility variation in monthly reporting for each indicator.

A value is classified as an outlier if it meets EITHER criterion:

1. The value exceeds 10 times the Median Absolute Deviation (MAD) from the facility's monthly median for that indicator, OR
2. The value represents more than 80% of the total volume for a given facility, indicator, and time period

AND the reported count exceeds 100.

Outlier illustration

Health Centre B - Malaria diagnostic tests:

Month	Tests reported	Classification
January	245	Within expected range
February	267	Within expected range
March	**2,890**	**Outlier**
April	256	Within expected range

Probable cause: Data entry error (e.g., "2890" entered instead of "289")

Analytical impact: Without adjustment, the data would indicate an erroneous increase in malaria testing during March.

Outlier prevalence: FASTR output

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

Assessing relationships between indicators

Related health services demonstrate predictable relationships that can be used to assess data quality. FASTR evaluates the following indicator pairs:

Indicator pair	Expected relationship	Rationale
ANC1 / ANC4	ANC1 ≥ ANC4	More women initiate antenatal care than complete four visits
Penta1 / Penta3	Penta1 ≥ Penta3	More children receive the first dose than complete the series
BCG / Deliveries	BCG ≈ Deliveries	BCG is administered at birth; counts should be similar

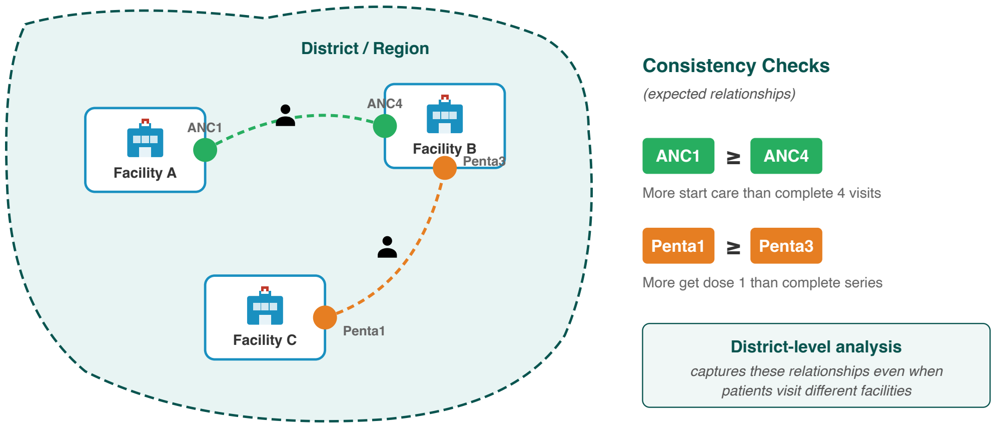
Violations of these expected relationships indicate potential data quality issues requiring investigation.

Why assess consistency at district level?

Patients often access different services at different facilities within a district:

- A woman may attend **ANC1** at a nearby health post, but travel to a health centre for **ANC4**
- A child may receive **Penta1** at a local clinic, but complete **Penta3** at a district hospital

Checking consistency at the facility level would miss these patterns. Aggregating to district level captures the complete picture of service utilization within a geographic area.



Internal consistency: FASTR output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

90% or above
80% to 89%
Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughy the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

Calculating the overall quality score

The composite score integrates all three data quality dimensions:

1. **Completeness:** Did the facility submit a report?
2. **Outlier status:** Are reported values within plausible ranges?
3. **Consistency:** Do related indicators demonstrate expected relationships?

Binary DQA score:

- Score = 1 if all three criteria are satisfied
- Score = 0 if any criterion is not met

Mean DQA score: Weighted average of completeness-outlier score and consistency score

Applications:

- Inform decisions regarding data inclusion in analyses
- Identify facilities requiring targeted data quality support

Overall DQA score: FASTR output

Overall DQA score

Percentage of facility-months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

80% or above
70% to 79%
Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

Mean DQA score: FASTR output

Mean DQA score

Average data quality score across facility-months

	2022	2023	2024	2025
Region 001	90.2%	95.5%	95.5%	96.5%
Region 002	87.3%	89.0%	88.2%	85.6%
Region 003	92.8%	94.8%	91.0%	84.3%
Region 004	88.0%	93.9%	93.0%	97.0%
Region 005	89.5%	93.6%	91.1%	88.6%
Region 006	93.2%	96.3%	93.4%	93.4%

80% or above
70% to 79%
Below 70%

Items included in the DQA score include: No missing data for 1) OPD, 2) Penta1, and 3) ANC1, where available; No outliers for 4) OPD, 5) Penta1, and 6) ANC1, where available; Consistent reporting between 7) Penta1/Penta3, 8) ANC1/ANC4, 9)BCG/Delivery, where available.

Rationale for data quality adjustment

Routine HMIS data contain two common limitations that can distort analytical results:

Issue	Impact on analysis
Outliers	Extreme values create artificial spikes in service volumes
Incomplete reporting	Missing data creates artificial declines that do not reflect actual service delivery

FASTR addresses these limitations by replacing problematic values with estimates derived from each facility's historical reporting patterns.

Adjustment scenarios

To support transparency and sensitivity analysis, FASTR produces four parallel datasets:

Scenario	Description
Unadjusted	Original reported values
Outliers adjusted	Extreme values replaced
Completeness adjusted	Missing values imputed
Both adjusted	All corrections applied

Indicators excluded from adjustment

Certain indicators are excluded from the adjustment process:

- **Mortality indicators** (maternal deaths, neonatal deaths, under-5 deaths): These represent discrete events where smoothing or imputation is not appropriate
- **Low-volume indicators**: Indicators that never exceed 100 reported events in any month are excluded from outlier adjustment

Service utilization analysis

Understanding how health service volumes are changing over time.

Two questions we answer

1. How is service volume changing?

Compare volumes year-over-year to identify increases or decreases across regions and indicators.

2. Are there disruptions from expected patterns?

Compare actual volumes to what we would expect based on historical trends, to detect and quantify shortfalls or surpluses.

Measuring change over time

We calculate **year-over-year percent change** to identify meaningful shifts in service delivery.

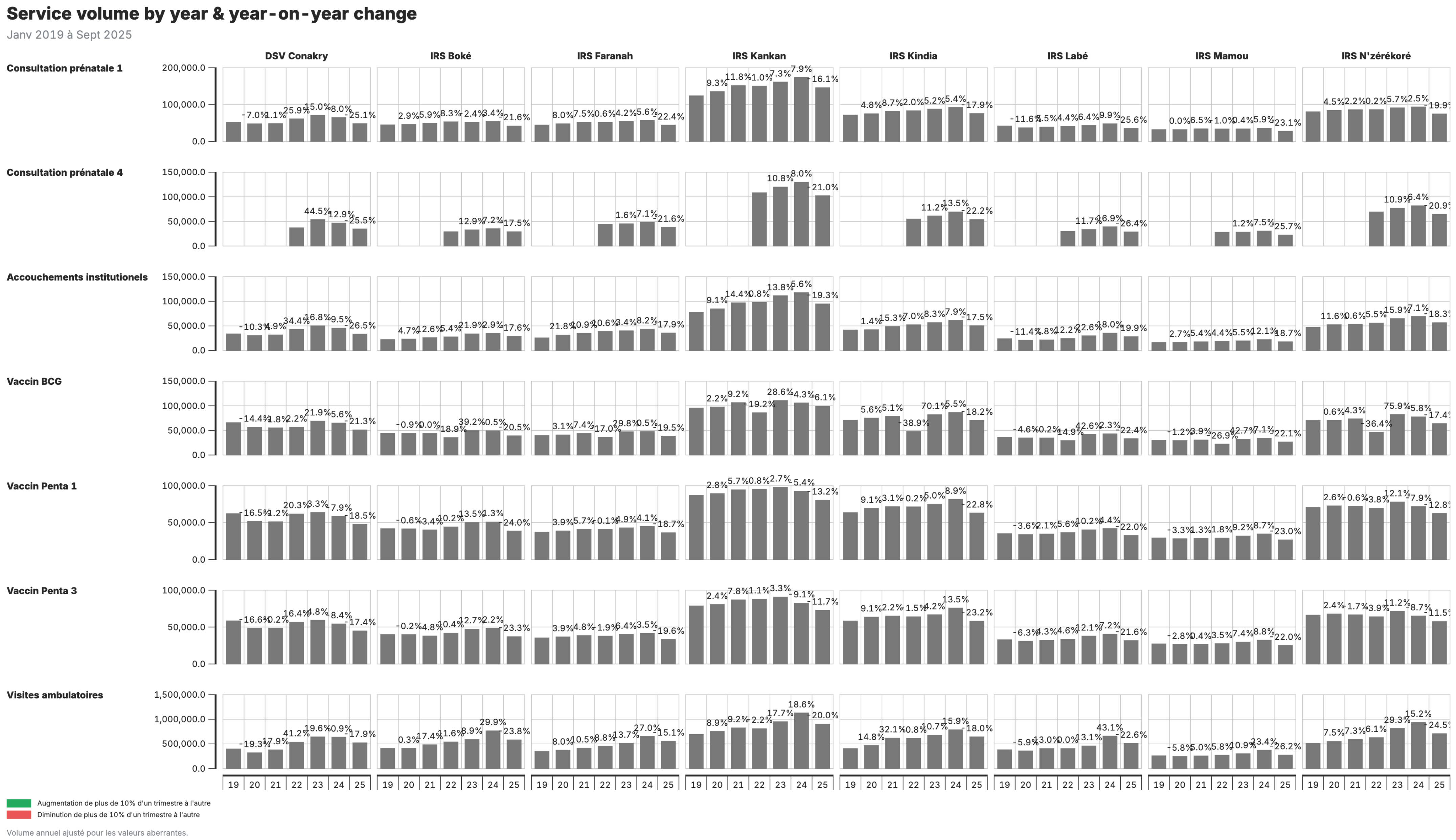
How it works:

For each indicator and region, we compare total volume in the current year to the previous year:

$$\text{Percent change} = \frac{\text{Current year} - \text{Previous year}}{\text{Previous year}} \times 100$$

Changes greater than **10%** (increase or decrease) are flagged for review.

Output: Change in service volume



Reading this chart

Bars show annual service volumes by region.

Percentages above bars show year-over-year change.

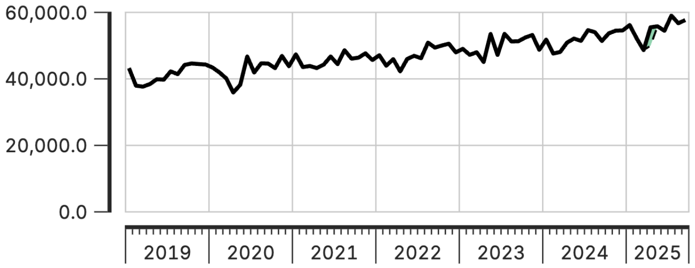
What to look for:

- Which regions show the largest increases or decreases?
- Are changes consistent across regions or localized?
- Do patterns differ by indicator?

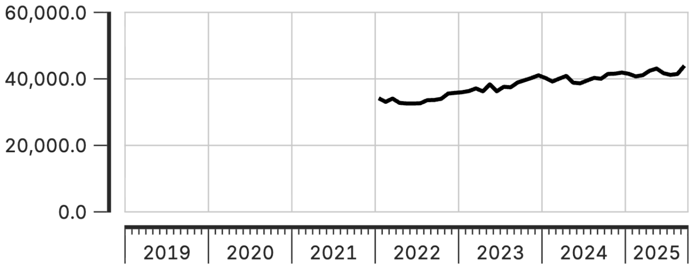
Perturbations et excédents du volume de service, au niveau national

Janv 2019 à Sept 2025

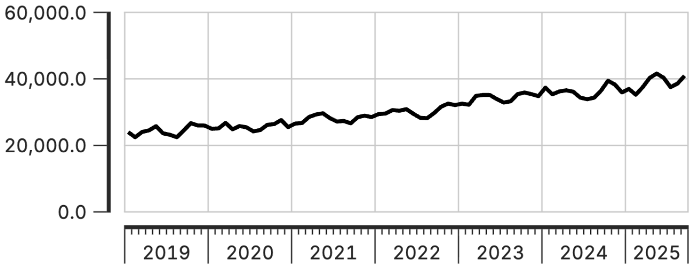
Consultation prénatale 1



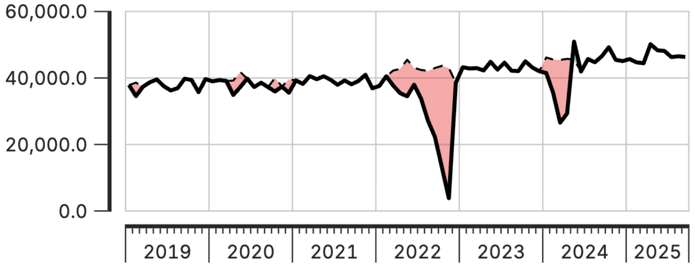
Consultation prénatale 4



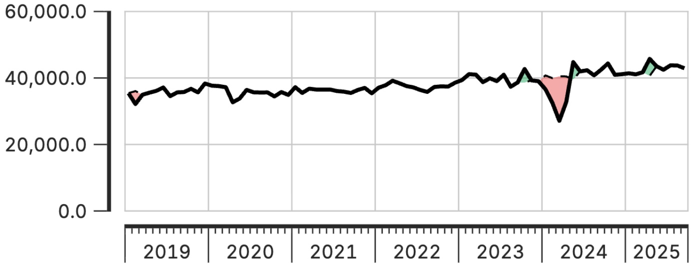
Accouchements institutionels



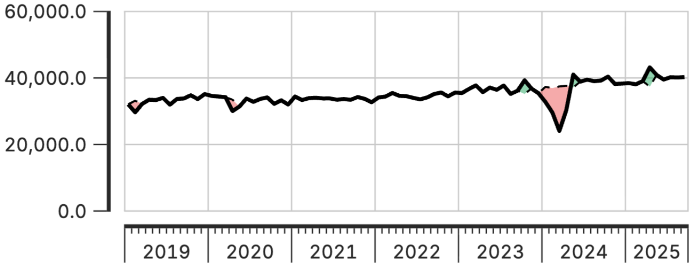
Vaccin BCG



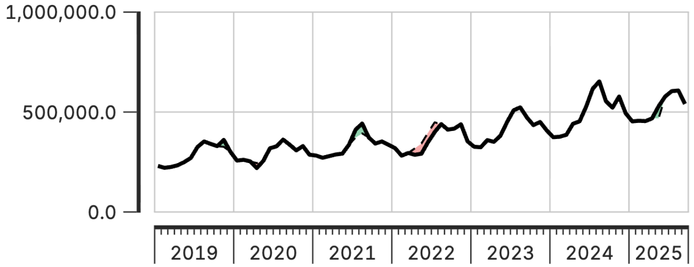
Vaccin Penta 1



Vaccin Penta 3



Visites ambulatoires

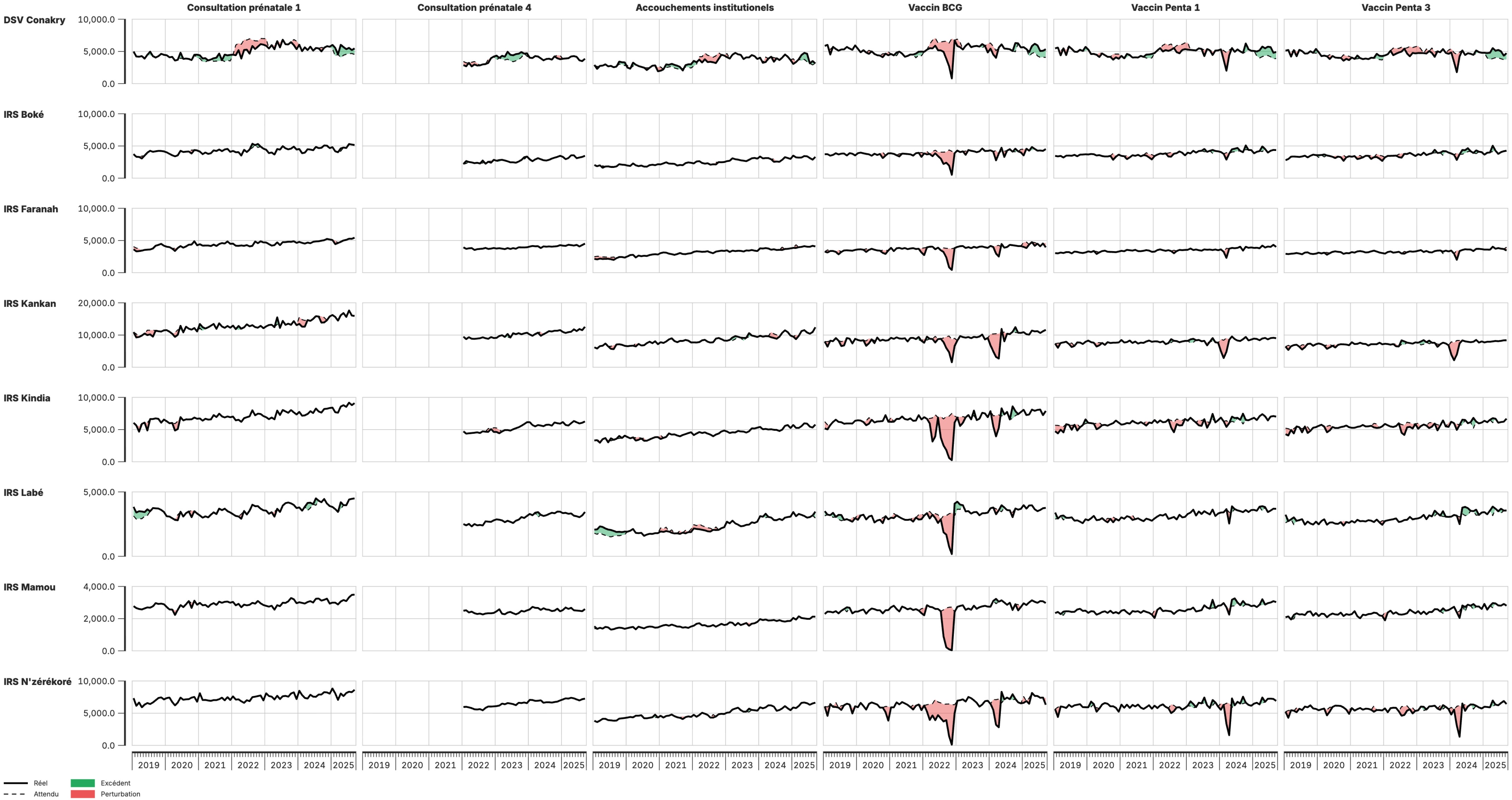


Réel
Attendu
Excédent
Perturbation

Ce graphique quantifie les variations au niveau du volume de services, par rapport aux tendances historiques, en tenant compte de la saisonnalité. Ces signaux doivent être triangulés avec d'autres données et connaissances contextuelles, pour déterminer s'ils résultent d'une qualité de données inadéquate. Les variations inattendues de volume sont estimées en comparant le volume observé au volume attendu, selon les tendances historiques et la saisonnalité. Les variations inattendues importantes dans les données historiques sont exclues. Cette analyse repose sur une régression de séries chronologiques interrompues avec effets fixes au niveau des établissements.

Perturbations et excédents du volume de service, au niveau infranational

Janv 2019 à Sept 2025



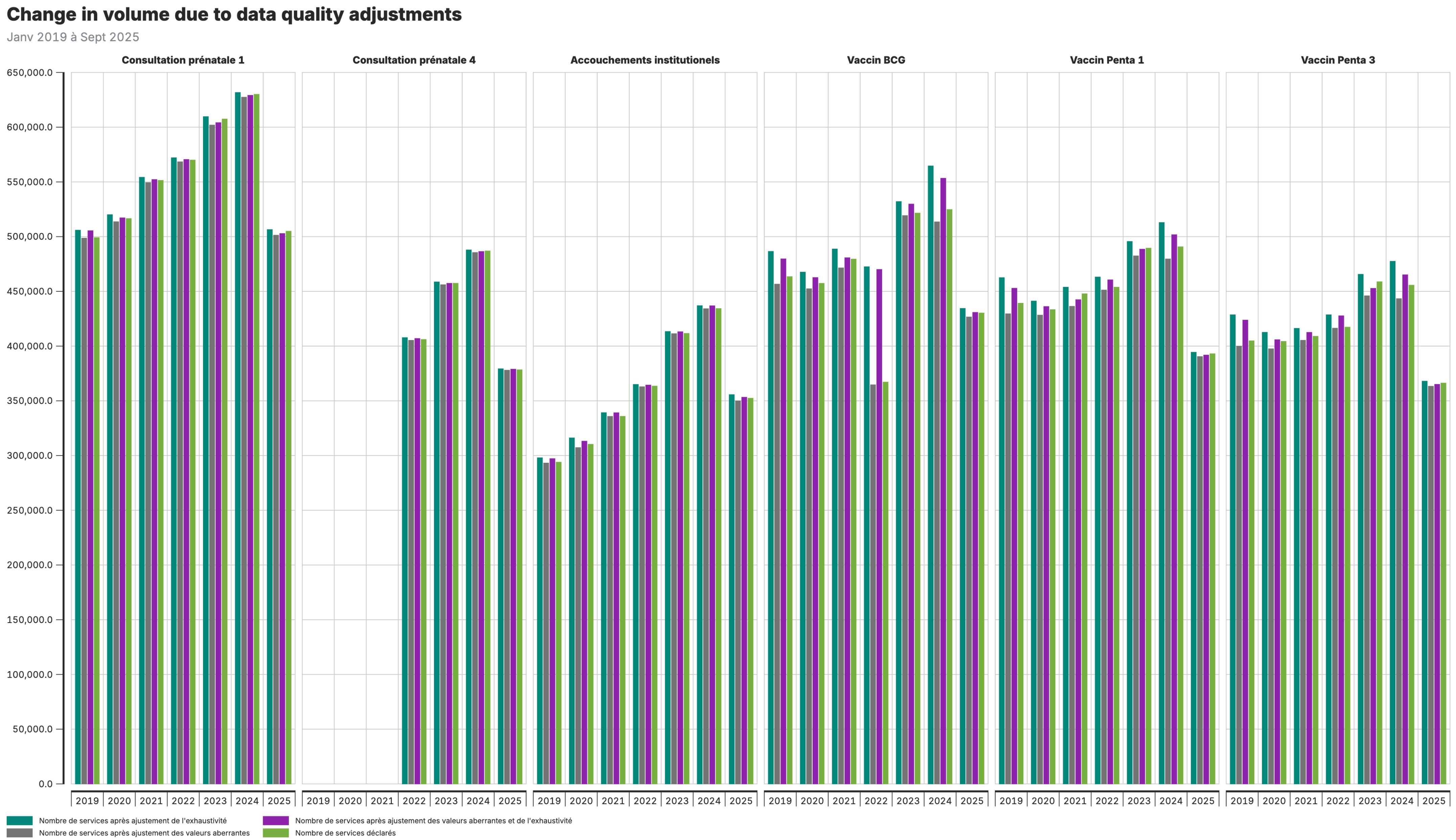
Ce graphique quantifie les variations au niveau du volume de services, par rapport aux tendances historiques, en tenant compte de la saisonnalité. Ces signaux doivent être triangulés avec d'autres données et connaissances contextuelles, pour déterminer s'ils résultent d'une qualité de données inadéquate. Les variations inattendues de volume sont estimées en comparant le volume observé au volume attendu, selon les tendances historiques et la saisonnalité. Les variations inattendues importantes dans les données historiques sont exclues. Cette analyse repose sur une régression de séries chronologiques interrompues avec effets fixes au niveau des établissements.

Impact of data quality adjustments

The analysis can use different versions of the data:

Scenario	What it uses
No adjustment	Raw reported values
Outlier adjustment	Extreme values corrected
Completeness adjustment	Adjusted for missing reports
Both adjustments	Outliers corrected + completeness adjusted

Output: Volume change by adjustment scenario



Why compare scenarios?

Comparing results across adjustment scenarios helps assess:

- **How much do adjustments change the picture?** If results are similar, findings are robust.
- **Which adjustment has the biggest impact?** Completeness vs outlier corrections.
- **Are conclusions sensitive to data quality assumptions?** Important for interpretation confidence.

Service coverage estimates

The Coverage Estimates module (Module 4 in the FASTR analytics platform) estimates health service coverage by answering: **"What percentage of the target population received this health service?"**

Three data sources integrated:

1. Adjusted health service volumes from HMIS
2. Population projections from United Nations
3. Household survey data from MICS/DHS

Two-part process

Part 1: Denominator calculation

- Calculate target populations using multiple methods (HMIS-based and population-based)
- Compare against survey benchmarks
- Automatically select best denominator for each indicator

Part 2: Coverage estimation

- Override automatic selections based on programmatic knowledge
- Project survey estimates forward using HMIS trends
- Generate final coverage estimates

What is service coverage?

Coverage answers: *What percentage of the target population received this health service?*

$$\text{Service coverage} = \frac{\text{Population who received the service}}{\text{Population who need the service (target population)}}$$

Numerator comes from DHIS2 data, count of services adjusted for outliers

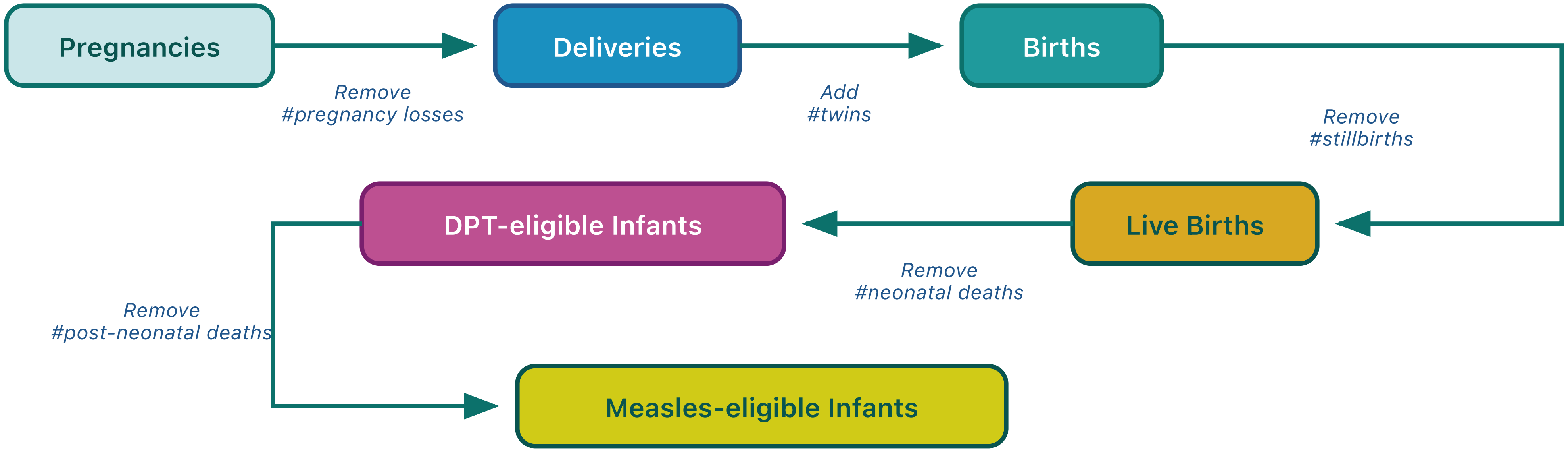
Several options for estimating the denominator

Types of denominators for FASTR core analysis

Type of service	Denominator
ANC	Pregnancies
Delivery	Live births
BCG	Live births
Penta1	Infants eligible for Penta (infants surviving 1+ months)
Penta3	Infants eligible for Penta (infants surviving 1+ months)

Expected relationships which help with estimating denominators

Starting from pregnancies, apply demographic factors to estimate other denominators:



Default values:

Pregnancy loss rate = 0.03 | Twin rate = 0.015 | Stillbirth rate = 0.02 | NMR = 0.03 | PNMR = 0.02

Use country-specific values for stillbirth rate, NMR, PNMR, and IMR when available from DHS/MICS.

Estimating denominators from ANC-1

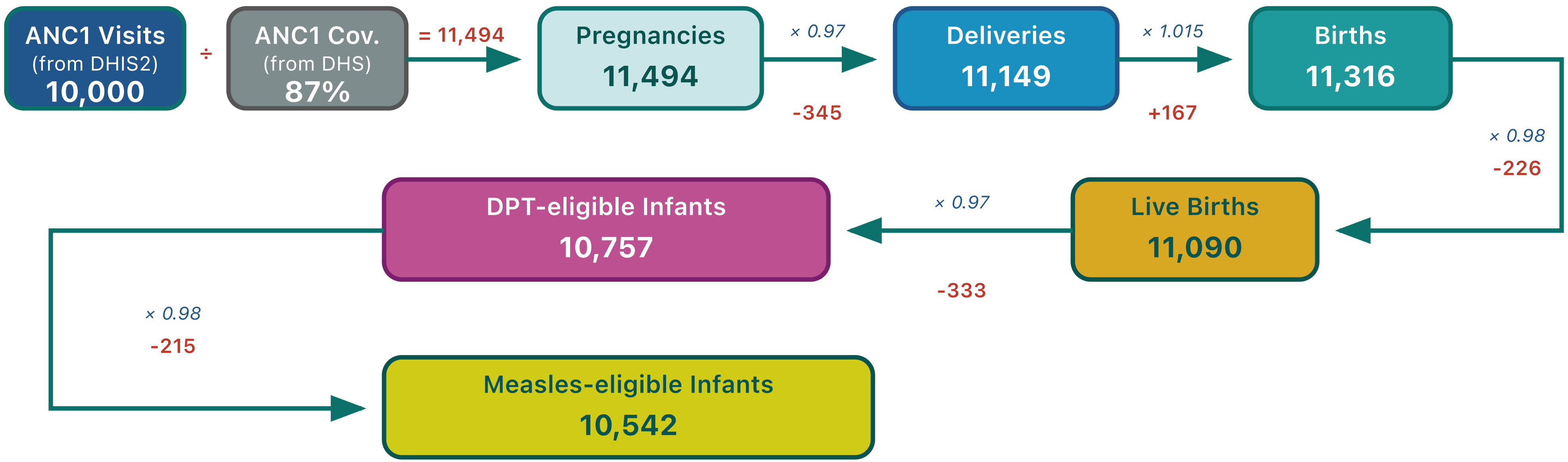
Using survey coverage + DHIS2 counts to derive denominators:

Step	Formula	Example
Pregnancies	ANC1 count ÷ ANC1 coverage	100,000 ÷ 0.95 = 105,263
Deliveries	Pregnancies × (1 - stillbirth rate)	105,263 × 0.97 = 102,105
Live births	Deliveries × survival rate	102,105 × 0.98 = 100,063

Estimating denominators from ANC-1: Visual example

Starting from ANC1 visits, apply demographic adjustment factors to estimate denominators for other services:

Example: Calculating denominators from ANC1 visits



Calculation summary:

Step 1: Pregnancies = ANC1 visits ÷ ANC1 coverage = 10,000 ÷ 0.87 = 11,494

Step 2: Apply demographic factors through the cascade:

Pregnancies (11,494) → Deliveries (11,149) → Births (11,316) → Live Births (11,090) → DPT-eligible (10,757) → Measles-eligible (10,542)

Rates: Pregnancy loss = 3%, Twin rate = 1.5%, Stillbirth = 2%, NMR = 3%, PNMR = 2%

Multiple denominator options

Each HMIS indicator provides an **entry point** into the demographic cascade. From that starting point, the module calculates forward and backward to derive all target populations:

Entry point	Calculation	Denominators derived
ANC1	ANC1 ÷ coverage → Pregnancies	→ Deliveries → Births → Live births → DPT-eligible → Measles-eligible
Deliveries	Deliveries ÷ coverage → Live births	← Pregnancies ← ... and → DPT-eligible → Measles-eligible
BCG	BCG ÷ coverage → Live births	← Pregnancies ← ... and → DPT-eligible → Measles-eligible
Penta1	Penta1 ÷ coverage → DPT-eligible	→ Measles-eligible
UN WPP	Population projections	Pregnancies, Live births, DPT-eligible, Measles-eligible

The module calculates coverage using **each** denominator option, then selects the best one.

Selecting the best denominator

How the module selects the best denominator:

1. **Calculate coverage** using each denominator option
2. **Compare to survey** (DHS/MICS) benchmark for that indicator
3. **Select the denominator** that produces coverage closest to survey

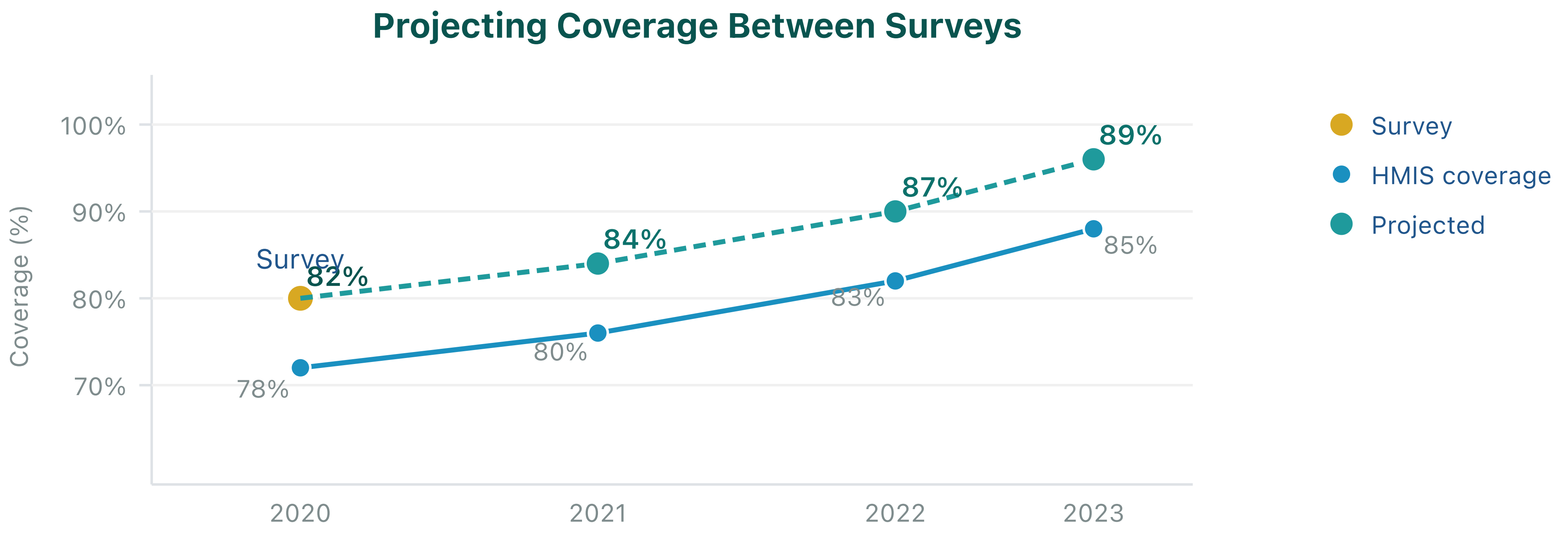
Selection rules

- **HMIS-based denominators preferred** over population projections
- **Independent denominators preferred** (e.g., using ANC1 to estimate Penta3 coverage, not Penta1)
- **Lowest error wins** when multiple options are available

This ensures coverage estimates are grounded in observed service delivery patterns.

Projecting coverage between surveys

Surveys (DHS/MICS) occur every 3-5 years. The module projects the last survey value forward using the trend observed in HMIS-calculated coverage:



How it works: Calculate the year-over-year change (delta) in HMIS coverage, then add each delta to the last survey value. This carries forward the survey baseline while incorporating observed HMIS trends.

Interpreting coverage outputs

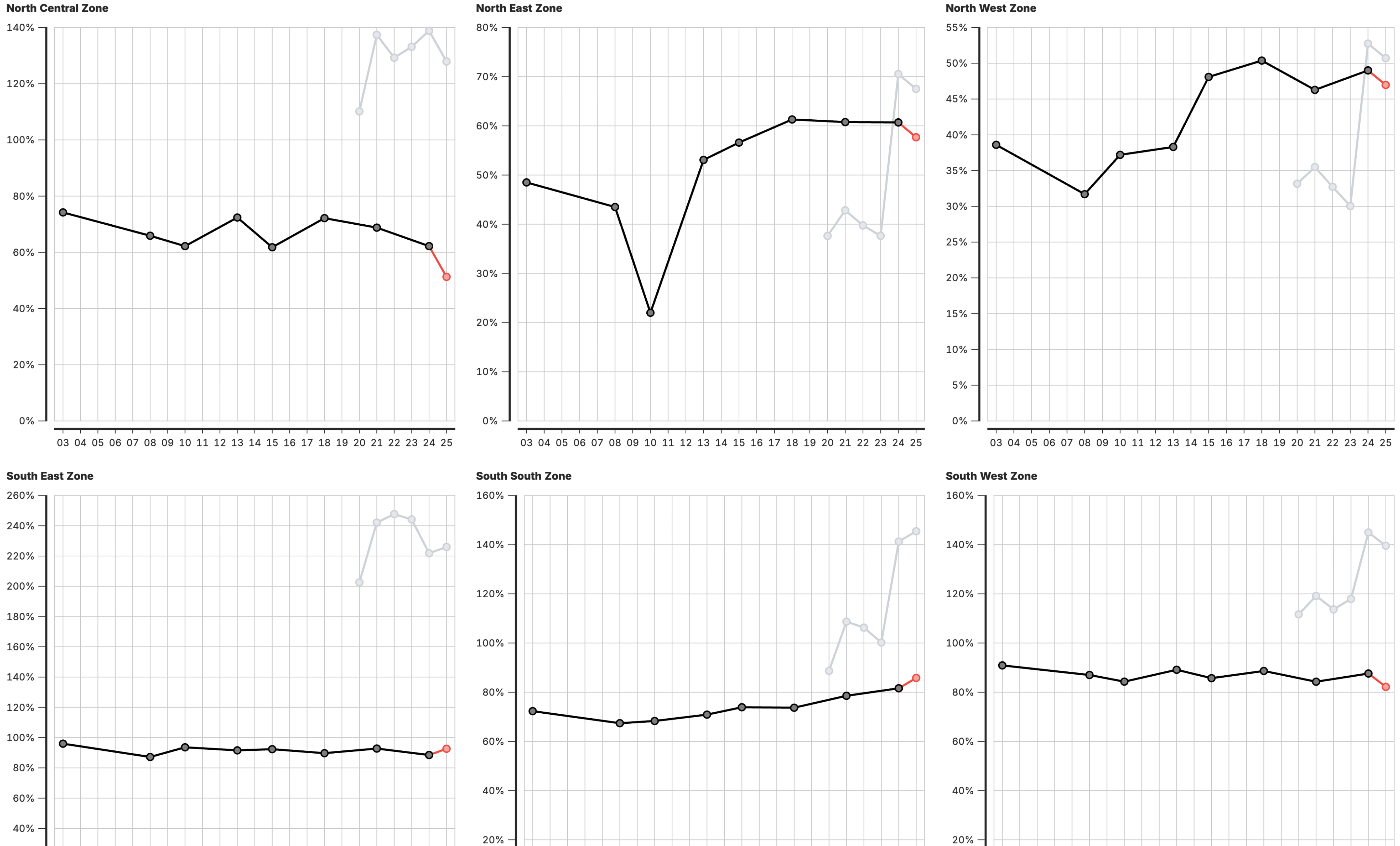
Understanding the chart lines:

- **Black line/points**: Survey data (DHS/MICS) - the reference standard from household surveys
- **Grey line/points**: HMIS-based coverage - calculated from facility data and selected denominator
- **Red line/points**: Projected coverage - survey estimates extended using HMIS trends

Key questions when reviewing

1. **How close is HMIS coverage to survey?** Large gaps may indicate denominator or data quality issues
2. **Are trends consistent?** HMIS and survey should generally move in the same direction
3. **Are projections plausible?** Coverage should stay between 0-100% and align with program knowledge

2. Coverage calculated from HMIS data (admin area 2)



Session 6: Creating a project

Activity: Creating a Project

In this hands-on session, we will:

- Set up a new project
- Configure project settings
- Select indicators and time periods
- Apply best practices for project organization

Participants will create their first project

Session 7: Creating visualizations

Activity: Creating Visualizations

In this hands-on session, we will:

- Explore available chart types
- Create and customize visualizations
- Export charts for use in reports

Participants will build visualizations from their analysis

🍽️ Lunch Break

90 minutes

Back at 14:00

Session 8: Creating reports

Activity: Creating Reports

In this hands-on session, we will:

- Use report templates
- Generate automated reports
- Customize report content and layout

Participants will create their first quarterly report draft

Day 3

Recap

On Day 2, we covered:

- FASTR methods for data quality assessment and adjustment
- Creating and configuring projects in the platform
- Building visualizations from Zambia data
- Generating reports

Day 3 Focus

Today we move from analysis to action:

- Deep dive into interpretation of results
- Create the Q4 2025 quarterly report
- Present findings to the group
- Develop action plans for continued work

Session 8: Interpretation of visualizations

Analytical thinking & interpretation

Content to be developed

This section will cover:

- Frameworks for interpreting FASTR outputs
- Connecting data patterns to programmatic meaning
- Common interpretation pitfalls to avoid
- Building analytical thinking skills

Session 9: Creating a Q4 2025 report

Generating quarterly reporting products

Content to be developed

This section will cover:

- Quarterly reporting workflow
- Using the FASTR platform for automated reports
- Quality assurance for reports
- Distribution and feedback mechanisms

🍽️ Lunch Break

90 minutes

Back at 14:00

Session 10: Presenting reports

End user mapping

End user mapping helps ensure that our outputs will meet the real needs of our end users.

Key questions

1. Who is my end user?
2. What does this end user need to accomplish with the report?
3. What information are they most interested in?
4. What do they like/not like about current reports?
5. How do they like to receive their information?

Day 4

Recap

On Day 3, we covered:

- Interpretation of FASTR visualizations
- Creation of the Q4 2025 quarterly report
- Presentation of findings and group feedback
- Action planning for continued platform use

Day 4 Focus

Today we design the Health Facility Assessment:

- Overview of the FASTR HFA phone survey
- Review questionnaire structure
- Adapt questionnaire for Zambia context
- Plan HFA priorities and data use

Session 12: Overview of FASTR HFA phone survey

Health Facility Assessment Phone Survey

Placeholder: Content to be developed

Lunch Break

90 minutes

Back at 14:00

Next Steps & Action Planning

Key actions to take after this working session:

- Finalize adapted phone survey questionnaire
- Complete first quarterly report draft
- Train remaining MoH team on data downloader tools
- Prepare roadmap for participation in Abuja workshop

Quarterly Reporting Cycle

Establish a sustainable reporting rhythm:

- Define quarterly report timeline and responsibilities
- Set up data extraction schedule
- Establish review and approval process
- Plan dissemination to stakeholders

Continued Platform Usage

Maintain momentum with the FASTR platform:

- Regular data updates and quality checks
- Expanding analysis to additional indicators
- Building capacity across MoH teams
- Documentation of lessons learned

Workshop Closing

Thank you for your participation!

Over the past four days, we have:

- Configured the FASTR platform for Zambia
- Created the first quarterly RMNCAH-N report
- Adapted the HFA questionnaire for local context
- Built capacity for ongoing analysis and reporting

Stay Connected

Continue the journey:

- Platform access and support
- Quarterly report submissions
- Multi-country workshop in Abuja
- Ongoing technical assistance from GFF team

Thank You

FASTR Working Session - Zambia
January 27-30, 2026
Lusaka

Contact Information

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